Danny Phantom starts dashing passing through walls at 15 m/s and after a while, its speed changes to 45 m/s. How many seconds have passed if his acceleration is 3 m/s^2?

Group of answer choices

9 s

8 s

7 s

10 s

Of the following situations, which one is impossible?

Group of answer choices

A body having velocity east and acceleration east

A body having constant acceleration and variable velocity

A body having zero velocity and non-zero acceleration

A body having constant velocity and variable acceleration

A body having velocity east and acceleration west

If an object is “released from rest”, what kinematic quantity does that tell you the value of?

Group of answer choices

The final velocity is zero.

The acceleration is zero.

The initial velocity is zero.

The time is zero.

If Heihachi Mishima vertically leaps reaching a height of 1.29 m, then what is his takeoff speed?

Group of answer choices

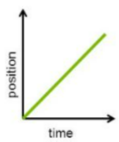
12.85 m/s

5.03 m/s

22.62 m/s

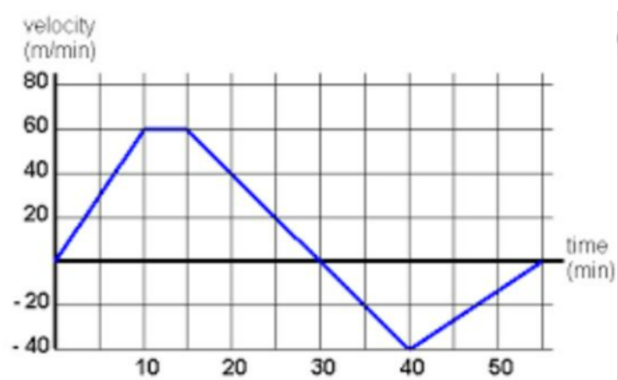
8.73 m/s

Question 9 1 pts



Which of the following is not true regarding the position vs. time graph shown above?

- ○ The object never changed its direction.
- ○ The object is moving forward.
- ◉ The object is at rest.
- ○ The object has a constant speed.



Given the velocity vs. time graph above describes the journey of a lion looking for his cub. At which time interval is the lion is speeding up?

- ○ 20 min to 25 min
- ○ 25 min to 30 min
- ◉ 30 min to 40 min
- ○ 0 min to 10 min

If an object is merely dropped (as opposed to being thrown) from an elevated height, then the initial velocity of the object is _________.

Group of answer choices

Zero

Positive

Negative

Infinite

During a soccer game, Abraham wants to kick the ball at an angle of inclination that will produce the maximum horizontal range. Which angle of inclination should he consider to attain his goal?

Group of answer choices

90 degrees

60 degrees

80 degrees

45 degrees

20 degrees

Ignoring air resistance, the horizontal component of the velocity a projectile motion is _________.

Group of answer choices

constant

increasing

decreasing

zero

Which of the following angles of inclination should one throw a rock to reach a maximum horizontal distance?

Group of answer choices

45 degrees

60 degrees

89 degrees

30 degrees

What happens to the centripetal acceleration of an object in uniform circular motion if its velocity doubles?

Group of answer choices

The centripetal acceleration will triple.

The centripetal acceleration will quadruple.

The centripetal acceleration will quintuple.

The centripetal acceleration will double.

An object has an initial velocity of 3 m/s after 10 seconds it stopped at 25 meters distance from its initial position. What is the final velocity of the object?

Group of answer choices

2 m/s

3 m/s

4 m/s

1 m/s